

Rai, Arpit

From: Desjardins, Raphael
Sent: 03 March 2026 02:40
To: Rai, Arpit
Cc: Raja, Ravisankar; Desrochers, Nicolas; Gowda, Murali; Jahan, Rameesa; Guan, Boyuan; Jaouani, Jihed
Subject: RE: Scarborough - NEW T1 Design Options - For Review

Thank you Arpit for this detailed email and for your good work. We will review what you are proposing as soon as possible and get back to you with our feedback.

In the meantime, do you know when we are expecting to receive the T0 track regression? And also the adjustment to the alignments and the blocks for the 4 existing crossovers following Nicolas's latest comment?

Best regards,

Raphaël Desjardins

Track Designer

TTA

AtkinsRéalis

On Corridor GO Rail Expansion Program

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AtkinsRéalis - Baseline / Référence

De : Rai, Arpit <Arpit.Rai@atkinsrealis.com>

Envoyé : 2 mars 2026 09:04

À : Guan, Boyuan <Boyuan.Guan@atkinsrealis.com>; Desjardins, Raphael <Raphael.Desjardins@atkinsrealis.com>

Cc : Raja, Ravisankar <Ravisankar.Raja@atkinsrealis.com>; Desrochers, Nicolas <Nicolas.Desrochers@atkinsrealis.com>; Gowda, Murali <Murali.Gowda@atkinsrealis.com>; Jahan, Rameesa <Rameesa.Jahan@atkinsrealis.com>

Objet : Scarborough - NEW T1 Design Options - For Review

Hi Boyuan/Raphael,

We have developed the design for the NEW T1 track, based on our discussion with Nicolas and the same has been demonstrated below for your reference:

Horizontal Alignment:

1. A track spacing report between BL 6.5 Alignment (LSE-PR-CA_ABCDE-T1-1) and EXS T1 (TTA-3710-SWS-ALG-ERM-T1.3) showed that there is insufficient Track centre buffer (<20mm) between CH. 1012+175m and CH. 1012+540m.
2. We have copied the above BL 6.5 Alignment to a new name: LSE-PR-CA_ABCDE-T1-1-MOD and nudged the straight section making it parallel to EXS T1 towards north to a maximum of 58mm which has allowed us to maintain a minimum Track centre buffer of 30mm at existing worse buffer location (CH. 1012+450m buffer = - 27mm)
3. These changes are local and do not affect the bearing of the Straight section at Brichmount Rd towards east end or Proposed Crossover towards west end.
4. The existing sub-ballast shoulder width based on existing ground surface in this area is wide enough to accommodate the proposed changes from BL 6.5.

Vertical Alignment:

1. The ballast depth in the BL 6.5 Profile (LSE-PR-CA_ABCDE-T1-1-TOR) showed inconsistency w.r.t the Ballast depth limits ($305 \leq b \leq 350$)
2. Option1: For discussion purposes, we began at CH. 1011+630m and optimised the ballast depth till CH. 1012+600m beyond which a large Parabola would connect to BL 6.5 grade of 0.38% at turnout location (PS CH. 1012+858.496m)
3. Option2: We copied Option1 and at east end, we started with grade of 2026-1 (1.407% @ CH. 1011+308.937m) and continued optimising the ballast depth till CH. 1012+600m beyond which a large Parabola (from CH. 1012+672.549m to CH. 1012+842.549m) would connect to a new grade of 0.368% provided by Nicolas based on track lowering trails at turnout location as per adjacent tracks elevations.
4. Option3: Option2 provided same number of number of vertical curves as BL 6.5 design. But when connecting the grades G1: 0.951% with G2: 0.368% (diff = 0.583%) a large curve had to be provided to bring vertical acceleration values to compliance and we could not optimise ballast depth as well. So we decided to prepare another option (option3) where the large curve is removed and a linear is inserted with 2 curves on either side which resulted in overall lesser lifts and passed vertical acceleration compliance in this area.

You can find the model containing the alignment in SKT folder on ACC here: [North Track Improvement.dwg](#)

Track spacing reports, geometry calculation sheets, sub-ballast shoulder check and Tie depths comparison: BL 6.5 vs

Option2/Option3 in the excel file:  [North Track Improvement.xlsx](#) placed on SharePoint.

Please let us know your comments on the design and improvements, we are happy to setup a call to discuss this in detail, if required.

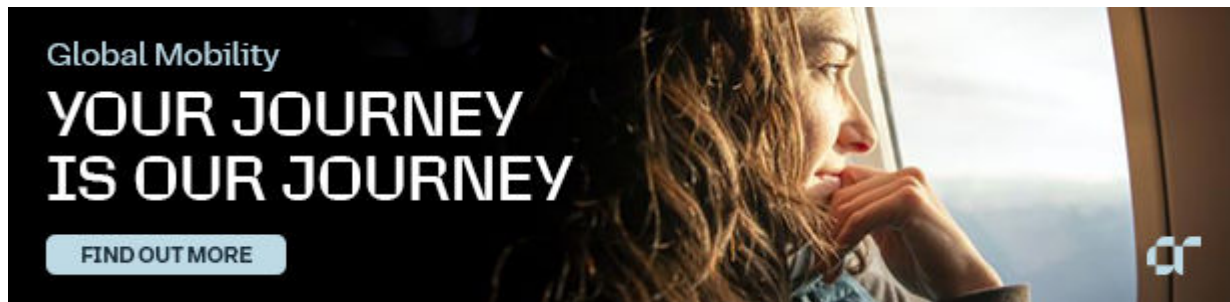
Many Thanks,

Arpit Rai (He/Him) B.E. (Civil)

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